

The Abelianized Rung and the T_k Classification

The two remaining links of the last volume, discharged. The abelianized rung reduces exactly to commutative Hankels conjugated by the square-rooted multinomial coefficients, and the transport fails off the axes by the closed-form ratio $\rho(\gamma) = \sqrt{\mu(\gamma)}$. Beyond T_0 the Hankel-preserving conditions collapse into one monoid homomorphism, and the isometry rigidifies the class to the multiplicative gradings on the weight sphere. The sandwich is measured honestly — closed, open, and strictly failed — by the multinomial budget.

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1 · The Mandate: Two Links, One Ladder

Volume VII closed with a ledger and a boundary. The seven demands of the full-proof order were discharged, the multiletter AAK equality was proved on the exact class of level-constant symbols, and the remaining links were named. Two of them were named as the next natural attacks: the abelianized class as the intermediate rung, where the transport was expected to fail by exactly the multinomial weights; and the classification of the Hankel-preserving isometries beyond the T_0 layer, which might rigidify the exact class to the geometric family. This volume is the execution of that order. Both links are now theorems with batteries, and the two answers turn out to be two views of one object: the multinomial measure of the Parikh lattice.

The shape of the answer is worth stating before the machinery. The abelianized symbols — those that see a word only through the counts of its letters — form a genuine intermediate rung between the level-constant class where Volume VII's transport succeeds and the full multiletter class where the problem is open. On this rung the transport fails for a reason that is now completely explicit: the Hankel operator of an abelianized symbol is an isometric copy of a commutative Hankel matrix conjugated by the diagonal of square-rooted multinomial coefficients, and that diagonal is precisely what the classical theory cannot absorb. Where the weights collapse — the level rung, where the fibre counts sum to n^k — the transport succeeds; where they are trivial — the coordinate axes, where every fibre is a single word — it succeeds again; and everywhere else it fails by the exact ratio $\rho(\gamma) = \sqrt{\mu(\gamma)}$, the square root of the multinomial coefficient of the cell.

The order's two items are one theorem seen from two sides: the transport fails by exactly the multinomial weights (item 1), and the admissible reweightings that survive every T_k are exactly the per-letter geometric family whose total squared weight is one — the multinomial budget (item 2). The isometry sphere of the classification is the boundary of the atom domain of the approximation problem.

The verification battery for this volume is `abelian_rung.py`, eight parts, with results in `abelian_rung_results.json`. Every identity claimed in the text is machine-checked: the Parikh reduction to $2.2e-16$ across six symbol families, the forced-weight lemma in both directions, the defect-ratio law on nine cells to $1e-12$, the spectral law with error exactly 0.0, the box determinants to machine precision, the golden-ratio witness on the anisotropic graded class to $5e-16$, the defect law of the rigidity induction to $2.8e-17$ over twelve random instances, and the strict-failure witness measured over a rank-1 family that is provably exhaustive. As in every volume of this programme, the honest boundary is drawn where the proofs stop, not where the optimism would like it.

2 · The Intermediate Rung: Parikh, Multinomials, and the Exact Reduction

Fix the alphabet S with n letters and let $m: S^* \rightarrow \mathbb{N}^n$ be the Parikh map, sending a word to the vector of its per-letter counts. A symbol h is abelianized when $h(w) = \phi(m(w))$ for some function ϕ on the lattice: the symbol sees the multiset of letters and nothing else. The level-constant class of Volume VII is the special case where ϕ depends only on the total degree, and the one-letter case is the degenerate case $n = 1$ where the Parikh map is the length. The fibre over a lattice point α — the set of words with those letter counts — contains $\mu(\alpha) = |\alpha|! / (\alpha_1! \dots \alpha_n!)$ words, the multinomial coefficient, and these counts are the measure of the rung.

The fibre structure gives a canonical isometry V_α from $\ell_2(\mathbb{N}^n)$ into $\ell_2(S^*)$, mapping the basis vector at α to the unit-normalized indicator of its fibre. The first theorem of the volume is the exact reduction: for every abelianized symbol, the multiletter Hankel operator H_h factors as $V_\alpha K V_\alpha^*$ with $K(\beta, \alpha) = \sqrt{\mu(\beta) \mu(\alpha)} \phi(\beta + \alpha)$. The matrix K is the commutative Hankel — the catalecticant of the form with coefficients ϕ — conjugated on both sides by the diagonal of square-rooted multinomials. The identity is proved by entrywise computation, and the battery verifies it to $2.2e-16$ on six symbol families: level-constant, the $(1,1)$ and $(2,1)$ cells, an axis-supported symbol, a multiplicative grading, and a random abelianized symbol. Singular values and rank are preserved exactly, as any isometric conjugation preserves them.

Two anchors make the reduction honest rather than merely true. First, the level rung of Volume VII is recovered inside it: lumping the lattice by total degree gives an isometry L with $V = V_\alpha L$, and for level-constant ϕ the transported matrix collapses to $L H_\psi L^*$ with $\psi(k) = \phi(k) n^{\lfloor k/2 \rfloor}$ — the exact formula of Volume VII, reproduced by the battery to $2.2e-16$. The collapse happens because the fibre counts sum by level: $\sum_{|\alpha|=k} \mu(\alpha)$ equals n^k , depending only on k . Second, the one-letter sanity: for $n = 1$ the multinomial weights are identically 1, the Parikh lattice is the degree axis, and the reduction says the multiletter Hankel is the classical one-letter Hankel — verified exactly, as it must be. The rung is genuinely intermediate: it contains the level class strictly, and is contained in the full class strictly, for every $n \geq 2$.

3 · The Weights Are Forced: The Parikh-Block Lemma

An approximation theory on the rung needs to know which operators on the Parikh side are themselves Hankel after transport. The answer is the second theorem, the Parikh-block lemma: an operator of the form $V_\alpha K' V_\alpha^*$ is a multiletter Hankel operator if and only if K' carries the same multinomial weights — that is, $K'(\beta, \alpha) = \sqrt{\mu(\beta) \mu(\alpha)} \psi(\beta + \alpha)$ for some ψ — in which case the transported operator is precisely the Hankel of the abelianized symbol ψ composed with m . The approximants on the rung are forced to live in the same weighted family as the target.

The proof of the forced direction is a cut argument. If the transported matrix is Hankel with symbol g , then for every word x the value $g(x)$ must agree across all cuts of x , and the entries across a cut depend only on the Parikh pair $(m(u), m(v))$ of the cut pieces. Every decomposition $\beta \leq \gamma$ is realized by a cut of some word — take any word of Parikh β and concatenate any word of Parikh $\gamma - \beta$ — so the value must in fact depend only on the sum $\beta + \alpha$. Unwinding the fibre normalization gives exactly the weighted catalectic form. The battery checks both directions: ten random weighted catalectics transport to Hankel operators (10/10), ten random block matrices fail (0/10), and the unweighted transport fails with a recorded conflict of exactly the multinomial ratio.

That recorded conflict is worth seeing in digits because it is the whole story in one entry pair. Transport the commutative Hankel without the weights and look at the word ab . The cut (ϵ, ab) produces the entry $\psi(1,1)$ divided by $\sqrt{\mu(0,0) \mu(1,1)} = 1.414$, while the cut (a, b) produces $\psi(1,1)$ divided by $\sqrt{\mu(1,0) \mu(0,1)} = 2.0$ — the same word, the same symbol value, two different transported entries, their ratio exactly $1/\sqrt{2}$, the reciprocal of the square root of the $(1,1)$ multinomial. The battery measures this pair at 1.4142 and 2.0 against the predicted 1.4142. The weights are not a technical inconvenience; they are the difference between a well-defined symbol and a contradiction, and any approximant that wants to compete on the rung must pay them in full.

The lemma has an immediate structural consequence: the structured approximation problem on the rung is exactly the problem of minimizing the operator norm of $D_\mu^{0.5} (\text{Cat}_\phi - \text{Cat}_\psi) D_\mu^{0.5}$ over symbols ψ whose catalectic has rank at most M . The rung inherits the commutative-multivariable Hankel approximation problem — with the multinomial diagonal attached. This is the precise sense in which the rung is intermediate: it is the commutative problem, reweighted by the measure of the lattice. Every closed and open question that follows in this volume is a statement about that diagonal.

4 · The Localization Theorem: Where the Transport Survives

The transport of the one-letter theory through V_α would need the transported matrix K to be a commutative Hankel itself — entries depending on $\beta + \alpha$ alone. The localization theorem says exactly when this happens: K is a commutative Hankel if and only if the multinomial weights are decomposition-constant on the support of ϕ , which happens if and only if the support is contained in the union of coordinate axes (with the origin). On the axes every fibre is a single word, $\mu = 1$, and the weights are trivial; off the axes they vary, and the variation is the failure.

The failure is measured by a closed form, and the closed form is the heart of the order's first item. For a cell γ , consider the profile of weights $\sqrt{\mu(\beta)} \mu(\gamma - \beta)$ as β ranges over the decompositions. The multivariate Vandermonde identity gives $\mu(\beta) \mu(\gamma - \beta) / \mu(\gamma) =$ the product of binomials $C(\gamma_i, \beta_i)$ divided by $C(|\gamma|, |\beta|)$, which is at most 1 — the denominator is the sum of the numerator over the whole degree slice — with equality exactly at the axis extremes. So the profile runs from 1 (at $\beta =$ the axis projection, where both fibres are single words) to $\sqrt{\mu(\gamma)}$ (at $\beta = 0$ or $\beta = \gamma$), and the transport defect ratio is $\rho(\gamma) = \sqrt{\mu(\gamma)}$, exactly, for every off-axis cell. The battery confirms this on nine cells to $1e-12$, including the near-axis cells like $(4,1)$ where the ratio is a modest 2.24 and the balanced cells where it grows without bound.

Cell γ	$\mu(\gamma)$	$\rho(\gamma) = \sqrt{\mu(\gamma)}$ exact	$2^k (\pi k)^{-1/4}$ asymptotic	ratio
(1,1)	2	1.41421	1.50225	0.94140
(2,1)	3	1.73205	-	-
(2,2)	6	2.44949	2.52648	0.96953
(3,1)	4	2.00000	-	-
(3,3)	20	4.47214	4.56586	0.97947
(4,4)	70	8.36660	8.49802	0.98454
(5,5)	252	15.87451	16.07385	0.98760

Table 2 — The transport-defect ratio, cell by cell: the measured strip spread against the closed form $\rho(\gamma) = \sqrt{\mu(\gamma)}$, and the balanced-cell asymptotic (exact to the shown digits; the ratio column converges to 1)

The growth law deserves its own sentence because it is the quantitative face of the obstruction. On balanced cells $\gamma = (k, k)$ the defect ratio is $\sqrt{C(2k, k)}$, which behaves like 2^k divided by the fourth root of πk : exponential in the degree, tempered only by a polynomial prefactor. The battery's

growth table shows the measured ratios against this asymptotic converging from 0.94 at $k = 1$ to 0.99 at $k = 5$. The transport does not merely fail off the axes; it fails worse and worse as the letters of a cell balance out, which is exactly the regime where the abelianized rung is most distinct from both the level rung and the axes. The multinomial mountain of Figure 1(a) is this profile for $\gamma = (6,6)$, running from 1 at the corners to 30.4 at the centre.

Between the two extremes the localization theorem leaves no gap. Where the support is axis-supported, K is a commutative Hankel — the battery measures the entry consistency defect at exactly 0.0 — and the rung's problem embeds in the commutative theory with trivial weights. Where any mass sits off-axis, the entry conflict on the gamma-strip is $\phi(\gamma)$ times $(\rho(\gamma) - 1)$, measured at 0.621 for the (1,1) cell loaded with 1.5, exactly as predicted. The transport survives precisely on the set where the multinomial measure is degenerate, and nowhere else on the rung. This is item one of the order, discharged in closed form.

5 · The Spectrum Is the Multinomial Profile

The single-cell abelianized symbols — $h(w) = c$ on the fibre of γ and zero elsewhere — have a completely explicit spectral theory, and it is the cleanest demonstration that the multinomial weights are not an artifact of the transport but the intrinsic geometry of the rung. For such a symbol the transported matrix K is a monomial matrix: one nonzero entry per row and per column, sitting at $(\beta, \gamma - \beta)$ with value c times $\sqrt{\mu(\beta) \mu(\gamma - \beta)}$. The singular values of a monomial matrix are the absolute values of its entries, so the spectrum of the cell is exactly the list of geometric means of complementary multinomial coefficients, scaled by c , and the rank is the number of decompositions, the product of the $(\gamma_i + 1)$.

The battery verifies the spectral law with error exactly 0.0 on four cells: the (1,1) cell has $\sigma = (\sqrt{2}, \sqrt{2}, 1, 1)$ and rank 4; the (2,1) cell has $\sigma = (2.2517, 2.2517, 1.8385, 1.8385, 1.3, 1.3)$ and rank 6; the (2,2) cell has nine singular values up to 1.7146 and rank 9; the (3,1) cell has eight and rank 8. A structural feature visible in every one of these lists is the pairing: the strip matrix is symmetric with zero diagonal blocks, so its eigenvalues come in plus-minus pairs and the spectrum is doubly occupied. The pairing has a sandwich consequence recorded in the witness chapter — at $M = 1$ the EYM floor equals the norm and the trivial approximant closes the gap — so the honest failure slots of a cell begin at $M = 2$.

The rank statement generalizes from cells to boxes, and the generalization is the register-inflation law. For a symbol supported in the downset box $[0, \gamma]$ with corner coefficient c nonzero, the box section of the catalectic has determinant plus-or-minus c raised to the box dimension — the reversal is the unique permutation of the box with $\beta + \pi(\beta) \leq \gamma$ for every β , forced layer by layer from the corner inward, so every other permutation of the determinant expansion hits a zero entry — and hence the catalectic rank is exactly the number of lattice points of the box, the product of the $(\gamma_i + 1)$. The battery computes these determinants on random box-supported symbols and recovers c^R to

machine precision, with rank R in every instance.

At the same total degree k , the level rung registers $k + 1$ and the abelianized rung registers the product of the $(\gamma_i + 1)$, which on balanced cells is quadratic in k : the (k, k) cell registers $(k+1)^2$ against the level's $2k + 1$, an inflation factor that grows linearly in k . The register — the minimal machine, the resource of the framework's rank-register clause — is inflated by the multinomial geometry itself. The abelianized ab/ba cell, with $h(ab) = h(ba) = 1$, registers 4; Volume VII's non-abelianized partial-realization witness, with $h(ab) = 1$ and $h(ba) = 0$, extends at register 2. Abelianizing the symbol is not the cheap direction: the symmetry constraint on the symbol side costs machine states, and the cost is again the multinomial count.

6 · The T_k Collapse: The Homomorphism Classification

The second ordered item concerns the structure theory of Volume VII: W , with columns w_k , is Hankel-preserving when $T_k =$ the sum of $w_i w_j^*$ over $i + j = k$ is a Hankel operator for every k , and Volume VII classified only the T_0 layer — w_0 multiplicative, the per-letter products with squared sum below one, plus the delta at the empty word. The question left open was what the conditions for $k \geq 1$ add. The answer begins with an observation that collapses the entire tower into a single algebraic condition: forming the column-series $F(x, z) =$ the sum of $w_k(x) z^k$, all the T_k conditions hold at once exactly when $F(x, z) F(y, z)$ depends on the pair (x, y) only through the concatenation xy — that is, exactly when x maps to $F(x, \cdot)$ as a monoid homomorphism from the free monoid into the formal power series under the Cauchy product.

Because the Cauchy product is commutative, any such homomorphism factors through the abelianization, and the homomorphisms of the lattice N^n are the per-letter powers: $F(x, z)$ is the product over letters of $f_a(z)^{m_a(x)}$, with f_a arbitrary per-letter formal series. This is the complete formal classification, and it is emphatically not rigid. The family is vast: the per-letter series may have constant terms, several terms, zero terms; the battery draws random multi-term series for both letters, builds the columns by polynomial arithmetic, and verifies the homomorphism identity on random word pairs (12/12) and the Hankel property of every T_k up to degree twelve (12/12). Random columns fail the same tests 12/12. T_0 's multiplicative classification is recovered as the $k = 0$ shadow — the constant term of the series — and nothing in the formal layer rigidifies anything.

The geometric family of Volume VII and the multiplicative gradings of this volume are the monomial sub-family: $f_a = c_a z$, a single term of degree one per letter. Within the monomial family the Gram matrix is diagonal with entries $(\sum |c_a|^2)^k$, so the family is a family of weighted gradings whose total weight $\rho = \sum c_a^2$ controls everything: $\rho < 1$ gives bounded contractions, $\rho = 1$ gives isometries, $\rho > 1$ gives unbounded operators. The reweightings of Volume VII are the isotropic points

of this family. The rigidity question — whether the conditions beyond T_0 force the class into something small — is therefore a question about what the isometry constraint does to the per-letter series, and that is the next chapter's theorem.

Layer (constraints imposed)	The resulting family	Rigid?	Battery
T_k Hankel for every k (the formal condition; Vol VII's S criterion, all k at once)	The per-letter series family $F(x,z) = \prod_a f_a(z)^{m_a(x)}$ with f_a ARBITRARY formal series (constants allowed; multi-term allowed; zero series allowed)	NO — rich	12/12 random f_a pass every T_k ; the homomorphism identity 12/12; random columns fail 12/12
+ 12 columns (boundedness of each w_k)	The same family, with the constant-term condition $\sum_a f_a(0) ^2 < 1$ for w_0 (the T_0 shadow; Vol VII's multiplicative classification recovered)	no	The w_0 norm identity $1/(1 - \sum_a f_a(0) ^2)$ via the multinomial generating function, partial sums exact
+ Gram = I (the isometry)	$f_a = c_a z$ EXACTLY (single term, degree 1) with $\sum_a c_a^2 = 1$: the multiplicative gradings — the per-letter anisotropic geometric family; complex unit λ are isometries but NOT Hankel-preserving (the $(a, \epsilon)/(\epsilon, a)$ cut pair sees λ_a vs its conjugate)	YES	The defect law 12/12 exact to 2.8e-17: $\ w_d\ ^2 = \rho^d + \sum_a z^d f_a^2$; the cross-term law 6/6; non-monomial breakage 12/12; five real gradings machine-exact (Gram, every T_k , transport, sigma)

Table 3 — The T_k classification, layer by layer: what each constraint adds, and what the battery measured

7 · The Isometric Rigidity: The Defect Law

Impose the isometry — the Gram matrix equal to the identity — on the per-letter series family, and the rigidity the order anticipated arrives. The proof is a degree induction driven by an exact formula, the defect law. Step zero: the norm of w_0 is one, and w_0 is the multiplicative function built from the constant terms $f_a(0)$; by the multinomial generating function its squared norm is 1 plus the geometric series of the total squared constant, which equals one only when every constant term vanishes. So $f_a(0) = 0$ for every letter: the columns are strictly positive-degree, and w_0 is exactly the delta at the empty word. Step one: w_1 is the vector of degree-one coefficients c_a on the single letters, and its unit norm fixes $\sum |c_a|^2 = 1$ — the live letters already carry the whole weight budget.

The induction step is where the higher coefficients die. Suppose all coefficients of every f_a vanish between degrees 2 and m . Then for k at most m the column w_k is exactly the multiplicative level vector — c to the Parikh exponent on the words of length k — and its norm is $\rho^k = 1$ as it must be. At degree $m + 1$ the column acquires exactly two kinds of new contributions: the level- $(m+1)$ main part, contributing ρ^{m+1} , and the single-letter contributions, the coefficients of z^{m+1} in each f_a , contributing their squared magnitudes — and nothing else, because the mixed corrections are pushed beyond degree $m+1$ by the induction hypothesis. The defect law is the resulting identity: the squared norm of w_{m+1} equals ρ^{m+1} plus the sum of the squared $(m+1)$ -th coefficients. The isometry

demands equality with $\rho^{m+1} = 1$, so every $(m+1)$ -th coefficient vanishes. Induction: $f_a = c_a z$, single term, degree one, and the weight budget is exactly spent.

The battery verifies the defect law directly, and it is the tightest measurement of the volume: twelve random instances with injected degree- d perturbations, the law holding with worst deviation $2.8e-17$. The single-letter contributions to w_d sit on level 1, where w_1 lives, so the Gram also acquires the exact cross-term w_1 dotted with w_d equal to the sum of $c_a \gamma_a$ — verified $6/6$ — a second, independent violation of the identity. Non-monomial series with two higher terms break the Gram in twelve of twelve random instances, and the monomial family with ρ not equal to 1 scales as ρ^k exactly. The rigidity theorem follows: the Hankel-preserving isometries are exactly the multiplicative gradings, $w_k(x) = 1[|x| = k]$ times the per-letter exponential of $m(x)$, with sum of the squared per-letter weights equal to one.

Two sharpenings complete the statement. First, the rigidity lands on the real sphere: complex unit λ give isometries but not Hankel preservation, because the cut pair (a, ϵ) and (ϵ, a) demands λ_a and its conjugate agree — the battery records the conflict at $|\operatorname{Im} \lambda_a| = 0.8$ for the tested instance. Second, the user's geometric-family guess is confirmed and refined: the isotropic point — all letters weighted equally, $c_a = 1/\sqrt{n}$ — is Volume VII's V , but the exact class is the whole per-letter sphere, and the anisotropy is genuine new freedom that T_k for $k \geq 1$ is precisely what pins down. The rigidification is to the multiplicative gradings, the per-letter geometric family, of which the isotropic grading is one point and the vertices — single live letters — are the axis-supported extremes.

8 · The Exact Class Assembled: Gradings, Vertices, and the Golden Witness

The rigidity theorem assembles the exact class. By Volume VII's W -generic clause — the sandwich transports through any Hankel-preserving isometry — and by the classification, the multiletter AAK equality holds on the multiplicative gradings of the level-constant symbols: $h(w) = \psi(|w|)$ times the product of $\lambda_a^{m_a(w)}$, with the per-letter weights on the real unit sphere. The transport is explicit: $W_\lambda H_\psi W_\lambda^*$ has entries $\lambda^{m(x)} \psi(|x| + |y|) \lambda^{m(y)} = \lambda^{m(xy)} \psi(|xy|)$, a Hankel operator whose symbol is exactly the graded one; the isometry preserves singular values; and the classical one-letter AAK approximant transports along with everything else.

The battery exercises the whole chain on five weight vectors — the anisotropic $(0.8, 0.6)$, the isotropic pair, a lopsided $(0.28, 0.96)$, a signed $(-0.6, 0.8)$, and $(0.6, 0.8)$ — and every link is machine-exact: the Gram at $8.9e-16$ or better, every T_k Hankel, the transport identity at 0.0 , singular values preserved to $4.4e-16$. The golden-ratio witness runs on the anisotropic grading: the graded symbol built from the one-letter golden symbol $(1, 1)$ has second singular value 0.6180339887 — the golden ratio conjugate,

exact to the last digit — and the transported rank-one Prony approximant, optimized at window six in the operator norm, attains 0.6180339887498945 against the floor 0.6180339887498949, a gap of $4e-16$. The approximant is verified Hankel with defect 0.0, of rank one, and its distance equals the one-letter distance through the isometry to $5.6e-17$.

The vertex gradings deserve their own paragraph because they repair an incomplete statement from the axis analysis. A vertex $\lambda = e_a$ concentrates the entire budget on one letter: the columns pick out the single words a^k , the isometry is trivial, and the transported class is the single-axis family — symbols supported on the powers of one letter, with the per-letter symbol arbitrary, not merely geometric. The battery runs the vertex transport end to end: the Gram and the transport identity at 0.0, every T_k Hankel, and the golden witness on the single-axis symbol attaining 0.6180339887498943 against the golden floor, with the isometry distance identity exact to $3.3e-16$. The single-axis class closes for arbitrary per-letter symbols because the vertex transport needs nothing geometric: the axis has one word per level, and the budget is spent on exactly that letter.

The pooled identity is the last structural piece: for genuine direct sums of one-letter Hankels, the $(M+1)$ -th pooled singular value equals the minimum over allocations of the maximum of the per-letter floors — the greedy allocation identity, verified exactly over M from 0 to 6 on a random pair of one-letter symbols. It is the mechanism behind any future multi-axis closure. But the two-axis symbols themselves are not direct sums: their Parikh-block matrices amalgamate over the origin — the shared empty-word row and column glue the per-letter blocks, and the amalgam of two rank-one per-letter blocks has rank three, not two. The battery records this corner tax explicitly. The corner is the first symptom of the budget obstruction, which the witness chapter turns into a measured strict failure.

9 · The Atoms and the Budget

The rank-one approximants on the rung have a complete classification, and it is the same classification again. A bounded weighted catalectic of rank one is an atom: $\psi'(\gamma) = p$ times the product of $\lambda_a^{\gamma_a}$, with the total squared weight strictly below one. The proof is the two-by-two-minor argument on the rank-one catalectic: the entries $u(\beta) v(\alpha)$ must depend on $\beta + \alpha$, which forces both factors to be exponential in the lattice and the symbol to be a per-letter power; the degenerate cases collapse to zero. The conjugated atom is the outer product of the weighted exponential vector $v(\beta) = \sqrt{\mu(\beta)} \lambda^\beta$ with itself, and the battery verifies the identity to $2.8e-17$ with rank exactly one on three instances.

The norm of the atom is where the multinomial generating function returns for the last time. The squared norm of v is the sum over the lattice of $\mu(\alpha)$ times the product of $|\lambda_a|^{2\alpha_a}$ — and the multinomial theorem collapses this to the geometric series of $\rho = \sum |\lambda_a|^2$, so the sum is $1/(1 - \rho)$ and the atom's norm is $|p|/(1 - \rho)$. The battery verifies the partial sums over the level-truncated lattice against the exact geometric partial sums to $1e-10$ and beyond, and the tail identity — the mass

beyond the box is exactly $\rho^{\{k+1\}}/(1 - \rho)$ — makes the truncation rigorous for the witness optimizations. Boundedness is exactly the condition $\rho < 1$: the atom domain is the open unit ball of the weight plane.

The domain's boundary is the isometry sphere of the rigidity theorem. The classification (Chapter 7) puts the exact transport class on the sphere $\sum \lambda_a^2 = 1$; the atom classification puts every rank-one approximant strictly inside it. The two parts of the order meet on this sphere, and Figure 1(b) draws the meeting: the interior is the approximation-theoretic regime, the circle is the transport-theoretic regime, the isotropic point is Volume VII's V, and the vertices are the single-axis extremes. One number governs both regimes — the total squared weight, the multinomial budget — and the next chapter shows a witness that fails precisely because it tries to overdraft it.

2.8e-17 worst deviation of the rigidity defect law, 12/12 instances	0.61803399 the golden witness on the anisotropic graded class, exact to 4e-16	1.0369 > 1 the two-golden amalgam: D(1) measured over the exhaustive rank-1 family
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10 · The Witnesses: The Sandwich's Fate on the Rung

Two witnesses calibrate the sandwich on the rung, one open and one strictly failed, and both are measured against floors that are proved and families that are exhaustive or honestly bounded. The first is the (1,1) cell itself: $h(ab) = h(ba) = 1$ and zero elsewhere. Its spectrum is $(\sqrt{2}, \sqrt{2}, 1, 1)$ — paired, as every cell's is — so at $M = 1$ the EYM floor equals the operator norm and the zero approximant closes the sandwich trivially; the battery records this closure explicitly rather than hiding it. The open slot is $M = 2$, where the floor is 1. The two-atom family — a provably valid family of rank-at-most-two approximants — measures 1.3188, beating the trivial bound of $\sqrt{2}$ but not the floor: the atoms' values on the cell's strip are tied to their geometric tails by the same multinomial weights, and no member of the family reaches down. The cell's D(2) lives in $[1.000, 1.319]$, and the exact value over all rank-two weighted catalectics — the polynomial-atom variants included — is open.

The second witness is the strict failure. Take the two-axis symbol with the golden one-letter symbol on each axis: $h(\epsilon s) = h(a) = h(b) = 1$ and zero elsewhere — the smallest symbol that is genuinely two-axis. Its transported matrix has spectrum $(2, 1, 0)$, so the EYM floor at $M = 1$ is exactly 1. The rank-at-most-one approximant family is exhaustive by the atom classification plus the zero, so D(1) is a three-parameter optimization, and the battery's twenty-four-start optimization converges to 1.0369. The floor is missed by 0.0369, and the miss is structural: matching the amalgam's first row would force $\lambda_a = \lambda_b = 1$, a total weight of $\rho = 2$, but the atom domain requires $\rho < 1$ and the isometry sphere sits at $\rho = 1$. The two-axis demand overdrafts the multinomial budget by the factor $n =$

2.

The sandwich fails on the rung for a reason that is now completely explicit: the multinomial budget. The admissible reweightings — the approximants inside, the transports on the sphere — all live under the budget sum $\lambda_a^2 \leq 1$, because the multinomial generating function sums their squared masses to the geometric series of the total weight. A symbol that wants n axes fully active asks for a total weight of n . For $n = 2$ the overdraft is witnessed: $D(1) = 1.0369$ against the floor 1, over the provably exhaustive rank-1 family.

The status of the two statements matters and is recorded carefully. The exhaustiveness of the rank-one family is proved — the atoms plus zero are all of it. The optimization over that family is numerical, multi-start, converged, with a rigorous truncation tail from the exact tail identity; the closed-form proof that the minimum exceeds the floor is not written here, and the ledger says so. What is proved is the structure: the failure, where it occurs, occurs because of the budget, and the budget is the multinomial measure. Figure 1(c) draws the calibration: the closed golden point on the graded class, the cell's open interval, and the amalgam's strict-failure interval, each with its floor marked.

11 · What This Changes: The Register, the Corpus, and the Boundary

For the corpus, the volume adds three theorems that stand on their own and connect outward. The exact reduction identifies the abelianized Hankel with the weighted catalecticant — the classical object of commutative algebra and invariant theory, the matrix of apolarity — reweighted by the multinomial measure; the rank-inflation law is the box-determinant lemma, an elementary but apparently new identity whose content is that the reversal is the unique surviving permutation; and the spectral law identifies the cell spectra with the geometric means of complementary multinomials. The commutative connection is recorded as context, honestly: the catalecticant literature is cited as the natural home of these matrices, and none of the volume's proofs depend on it.

For the framework's clauses, the volume sharpens two registers. The rank-register clause — resource equals rank — now carries the inflation law: the abelianized cell at multidegree γ registers the box count, polynomial of degree $n - 1$ in the total degree at balanced cells, against the level register's linear count; abelianization inflates the machine, and the multinomial count is the cost. The magnitude clause — the singular tail — now has the rung's measured floor structure: paired spectra for the cells, the multinomial profile as the spectrum itself, and the budget as the reason the profile cannot be matched cheaply. Both sharpenings are machine-checked and both are stated exactly where the boundary begins.

For the boundary itself, the volume replaces one large open problem with a smaller and sharper one. Volume VII's off-class boundary was the constructive nc-AAK problem in full generality. The rung analysis splits it: on the transportable families the equality is proved; on the off-axis rung the problem is now the multinomial-weighted catalectic AAK problem — approximate a weighted commutative Hankel by weighted commutative Hankels of bounded catalectic rank in the operator norm — with the EYM floor always available, the exhaustive rank-one family classified, and the first strict failure witnessed and attributed. The natural next links are named in the ledger: the closed-form minimum for the amalgam witness, the polynomial-atom families for the $M \geq 2$ slots, and the characterization of exactly which abelianized ϕ have the sandwich — the rung's own version of Open 7.13.

Item	Status after this session	Where
The abelianized reduction and the forced weights	CLOSED — the rung is exactly the multinomial-weighted catalectic problem; machine-exact	A3a, A3b
The transport failure, in closed form	CLOSED — $\rho(\gamma) = \sqrt{\mu(\gamma)}$ via the multivariate Vandermonde; exponential on balanced cells	A3c
The spectrum and the register of abelianized cells	CLOSED — $\sigma =$ the multinomial profile (paired); register = $\prod(\gamma_i + 1)$ (the box determinant)	A3e, A3f
The T_k classification beyond $k = 0$	CLOSED — the homomorphism classification (formal) + the isometric rigidity to the multiplicative gradings (the per-letter sphere)	S2a, S2b
The exact AAK class on the rung	CLOSED — the multiplicative gradings of the level-constant symbols; the golden witness machine-exact on the anisotropic family	A3d, S2c
The off-axis sandwich (the rung's own AAK problem)	OPEN — the multinomial-weighted catalectic AAK problem; the $\gamma = (1,1)$ cell: $D(2)$ in $[1, 1.319]$; the two-golden amalgam: $D(1) = 1.0369 > 1$ measured over the exhaustive rank-1 family (the budget overdraft); no closed-form minimum	A3h
The full classification of the rung's sandwich locus (which ϕ close exactly)	OPEN — the closed families found (the gradings, the vertices) and the witnessed failures bracket it; the characterization is the natural next link	outlook
The off-class equality of Vol VII (Open 7.13 / constructive nc-AAK), Φ 's uniqueness, Risk 4, the bounded benchmarks, the $n = 4$ $L = 10$ leg	UNCHANGED — carried forward	Vol VII ledger

Table 4 — The honest ledger: what this session closed, what it sharpened, and what it leaves open — the boundary carried explicitly

12 · The Honest Ledger

Both items of the order are discharged, and the discharge is asymmetric in the way the mathematics itself is asymmetric. Item one — the abelianized rung — is closed as a structure theory: the exact reduction, the forced weights, the localization, the two laws, the atoms, and the budget, with the sandwich's fate measured rather than claimed. Item two — the T_k classification — is closed in both layers: formally the homomorphism classification, which is rich rather than rigid, and isometrically the rigidity to the multiplicative gradings, which is the rigidity the order anticipated, refined from the isotropic geometric family to the full per-letter sphere. The junction is the theorem's shape: the transport's exact class and the approximation's rank-one class are the two regimes of one budget, and the budget is the multinomial measure of the Parikh lattice.

What remains is honest and bounded. On the rung: the weighted-catalectic AAK problem, with the cell's $M = 2$ slot open in $[1.000, 1.319]$ and the amalgam's $M = 1$ slot measured at 1.0369 against the floor 1 over the exhaustive rank-one family — the closed-form minimum, the higher-rank families, and the sandwich locus characterization are the named next attacks. Beyond the rung: Volume VII's ledger is unchanged — the off-class equality of Open 7.13, the uniqueness of the dictionary at infinite dimension, Risk 4 (the submission decision, which remains the author's alone), the bounded benchmark extensions, and the $n = 4$ $L = 10$ leg. The programme's architecture is now eight volumes deep, and every one of them ends the same way: with the proofs that exist, the measurements that bound what does not, and the names of the links that are next.